

Video Generation Prompt Optimization System

COMM7780 Final Project Report

Video Generation Prompt Optimization System

COMM7780 Final Project Report

GitHub Repository: <https://github.com/Linkaaa0102/COMM7780-Big-Data-Analytics-for-Media-and-Communication-Final-Project/tree/main>

1. Problem Definition

Current AI video-generation platforms (e.g., Jimeng, Seedance, Runway) require highly detailed and structurally specific prompts to produce stable, high-quality, and professional results. However, most end-users, such as content creators and digital artists, typically only provide simple, vague keywords or short descriptions (e.g., “A girl dancing on the beach”). This gap between user input and platform requirements leads to unpredictable generation outcomes, necessitating numerous trial-and-error attempts. This process significantly reduces creation efficiency, increases frustration, and wastes computational resources.

Core Problem: How to automatically and intelligently transform simple, non-technical user inputs into structured, professional-grade video generation prompts that are optimized for specific AI video platforms.

2. Dataset

Our system's knowledge base is constructed from authoritative and publicly available sources to ensure reliability and platform alignment.

Data Sources & Collection Method:

The initial dataset was **manually curated and organized** from the official documentation, user guides, and best practice publications of leading AI video generation platforms, including **Jimeng, Seedance, and Runway**. This method ensures the authority, accuracy, and legal compliance of our core data. We explicitly did not employ any unauthorized web scraping.

Data Composition:

The collected and organized knowledge encompasses:

Prompt Structures: Analysis of effective prompt templates and component sequences for different platforms.

Keyword Libraries: Categorized collections of descriptors for subjects, actions, scenes, lighting conditions, cinematic styles, and visual atmospheres.

Platform-Specific Rules: Adaptation rules and recommended parameters for different video generation engines.

Negative Prompt Compendium: A collection of common unwanted elements to improve output stability.

Data Storage:

The processed dataset is stored locally in two primary formats:

Structured Knowledge Files: Located in `/data/knowledge/`, these are organized text files (e.g., JSON, YAML) containing the categorized libraries and rules.

Application Database: Located in `/data/database/`, this SQLite database is used by the live system for efficient querying and retrieval during the prompt optimization process.

Raw Dataset: The original collected documents and the organized knowledge files are included in the project submission as the required raw dataset.

3. Training Process (Optional)

This project does not involve the training of a large-scale deep learning model from scratch. Instead, the system’s “intelligence” is built upon a **rule-based and knowledge-base-driven architecture**.

Core Methodology:

Sample Selection & Knowledge Organization: The foundational data was derived from the official platform documents mentioned in Section 2. This data was cleaned, categorized, and structured into a retrievable knowledge graph.

Labeling Rules & Enhancement Logic: We established clear rules for mapping user input keywords to relevant knowledge categories. For example,

the keyword “dancing” triggers the retrieval of action descriptors, camera movements suitable for dynamic scenes, and related style keywords.

Adopted Algorithms: The system employs a pipeline of **keyword analysis, semantic matching, and rule-based assembly**. It parses the user input, identifies key elements, retrieves relevant enhancement terms from the knowledge base, and assembles them according to a pre-defined, platform-adapted prompt template.

4. Model Application

The system acts as an intelligent middleware between the user and the AI video generation platform. The core “model” is the application logic that queries the knowledge base and constructs the prompt.

Application Workflow & Prompt Design:

A user interacts with a web interface. The key application step is the transformation of a simple input into a complex prompt.

User Input: The user provides a simple description (e.g., “A girl dancing on the beach”) and selects a target platform (e.g., Seedance) and style preference.

System Processing (The “Model” in Action): The system deconstructs the input, matches components to its knowledge base, and applies enhancement rules. It programmatically generates a detailed prompt structured as follows:

Enhanced Subject: “A young woman, detailed facial features, flowing dress,”

Enhanced Action & Scene: “elegantly dancing on a sunset tropical beach, smooth sand, gentle waves,”

Cinematic Language: “dynamic camera circling around the subject, slow motion,”

Lighting & Style: “golden hour lighting, cinematic, photorealistic, 4k.”

Negative Prompts: “Automatically appended from knowledge base: deformed, blurry, bad anatomy.”

Output: The final, optimized prompt is presented to the user for direct copy–and–paste into their chosen video generation platform.

5. Outcomes & Value

Solution Outcome:

The project successfully developed a fully functional, locally deployable web application that automatically converts vague user ideas into professional video prompts. The system demonstrably bridges the gap between novice input and professional output requirements.

Commercial & Social Public Value:

Lowered Barrier to Entry: Democratizes high–quality AIGC video creation by eliminating the need for deep expertise in prompt engineering.

Increased Productivity: Significantly reduces the time and effort spent on iterative generation attempts, streamlining the creative workflow for individuals and production teams.

Resource Efficiency: By improving output predictability and quality on the first attempt, the system can help reduce the computational waste associated with repeated, failed generations.

Educational Value: The system's knowledge base and enhancement process serve as a learning tool, helping users understand the components of an effective video generation prompt.

6. Additional Relevant Content

System Features:

Multi-Platform Support: Generates prompts tailored for Jimeng, Seedance, and general platforms.

Interactive Knowledge Base: Includes a interface for users to browse and learn from the curated prompt components and styles.

One-Click Local Deployment: Easy setup and execution via a provided batch script.

System Architecture:

Backend: Built with FastAPI (Python), SQLAlchemy, and SQLite.

Frontend: Built with Vue3, Vite, and Tailwind CSS.

The application runs locally on `http://localhost:8000`.

Usage:

Execute `启动系统.bat`.

Enter a creative description in the web interface.

Select the target platform and style.

Click “Generate Optimization Suggestion” to receive the final, refined prompt.